



ESSAY

Preventing a groundwater divide: Ensuring quality hydrogeology education for all in the age of online coursework following the COVID-19 pandemic

Matthew Currell^{1,2,3}

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Abstract

The COVID-19 pandemic forced an almost overnight rethink of traditional hydrogeology education, elevating online delivery as the primary means of teaching hydrogeology during 2020–2021. This change has had far-reaching effects, some of which persist in the form of greater reliance on online teaching formats in many universities. This paper describes the experience of delivering hydrogeology coursework in a city affected by the longest COVID lockdown in the world. Key lessons for hydrogeology education are examined.

Keywords Hydrogeology education · Groundwater and society · Groundwater education · Sustainability · Australia

Introduction

Overview and context

This article retrospectively examines the rapid and near-total transformation of hydrogeology coursework into fully online mode, which took place at an Australian university with a well-established hydrogeology curriculum, as a result of COVID-19. It explores how coursework normally delivered in-person was transformed into online teaching over a short period of time, and the challenges, opportunities and limitations of teaching hydrogeology in fully online mode. Since the pandemic, examples of partial or full online coursework remain at many institutions (e.g., modules delivered in ‘hybrid’ mode, with streamed or pre-recorded lectures followed by in-person classes) (e.g., Joshi et al. 2022; Newton and Draper 2025). These changes imply that COVID-19 precipitated (or accelerated) a major shift in methods of

hydrogeology education, in parallel with shifts occurring across the university sector (Detoni et al. 2025). Ensuring the benefits and weaknesses of this shift are carefully analyzed and understood, and applied to the future design of groundwater coursework is an important undertaking (Fischer and Tatomir 2022; Thompson et al. 2022). This is particularly true in the context of a push by many universities to increase the availability of courses to students studying remotely and/or with significant paid work or other commitments that clash with in-person classes (e.g., Joshi et al. 2022).

In the age of proliferation of online university coursework, resource-stretched education providers, and increasing efforts by universities to capture new audiences of students, key topics for discussion are whether online training is able to meet the needs of the groundwater profession, and whether it is suitable to equip the next generation of hydrogeologists with the knowledge and skills they need to solve the world’s groundwater challenges.

COVID-19 and the overnight online education transformation

The 2020–2021 COVID-19 pandemic was a global event that profoundly changed our society. The global public health response precipitated sudden, dramatic, and lasting effects on university education (Khawaja et al. 2023; Csorba and Dabija 2024). While online hydrogeology (and hydrology) education had already existed in various forms in the

✉ Matthew Currell
m.currell@griffith.edu.au

¹ School of Engineering & Built Environment, Griffith University, Nathan, QLD, Australia

² Australian Rivers Institute, Griffith University, Nathan, QLD, Australia

³ School of Engineering, RMIT University, Melbourne, Australia

years leading up to 2020, the pandemic forced a snap prohibition of traditional classroom, laboratory, and field-based hydrogeology education in most countries.

The city of Melbourne, Australia, was subject to the world's longest continuous 'lockdown', prohibiting the population from gathering for work or study (excluding exceptional cases), for a cumulative total of 262 days over 2020–2021. These public health restrictions forced university programs to reconfigure to a fully online environment, taking effect almost immediately, with only a few days of preparation time. As others have noted (e.g., Fischer and Tatomir 2022), most university teaching staff were ill-prepared and had little prior experience delivering fully online content and assessment. Many university staff (and students) were also challenged by the new working environment—teaching from home and fully dependent on a (periodically unstable) Internet connection, juggling children's education and other care responsibilities, and dealing with significant social isolation. In the years since, an increasing number of digital/online tools have become available, and their adoption (absent the pressures of the pandemic) has arguably improved elements of instruction (OECD 2023). However, it is also clear that such tools cannot replace careful preparation and curation of material and dedicated individualized student feedback (Joshi et al. 2022; OECD 2023).

The significant shift towards online university coursework triggered by the global COVID-19 pandemic has also had major ongoing consequences for education (Broadbent et al. 2024; Csorba and Dabija 2024). This includes evidence of changing student practices and expectations regarding the necessity of in-person attendance at classes (Detoni et al. 2025), and a push by many university managers to offer more online degree and course options.

In this context, it is vitally important to carefully analyze the strengths and limitations of fully online groundwater coursework. It should be noted that some institutions have opted to tighten requirements for in-person attendance for university classes, in recognition of potential negative impacts of virtual attendance on educational experience and achievement (Doran 2021).

It is hoped that the reflections and analysis below can help to identify where online learning can be utilized beneficially, e.g., complementing in-person learning, and also highlight where this may be detrimental.

The COVID-19 pandemic and its effect on hydrogeology teaching (2020–2021)

The forced transition to online hydrogeology teaching: Case study

Two hydrogeology coursework units (sometimes termed 'modules', or 'subjects' depending on the institution), with

between 20 and 55 students, were coordinated and taught by the author throughout 2020 and 2021 at RMIT University in Melbourne. During these 2 years, at least one offering of each unit was conducted in 100% online mode. Prior to the pandemic, they had been run entirely as face-to-face classes, involving a combination of lecture and tutorial sessions, and a field trip. The instructor was the primary teacher of both courses from 2013 to 2023.

The original shift to fully online delivery occurred just prior to the first week of the 2020 Australian academic year (starting in March). At the time, only a small amount of pre-existing online content (lecture recordings) was available. A decision was made not to use these recordings and instead deliver the new year's coursework utilizing livestreamed, recorded lectures and tutorial sessions, as well as new video content produced and designed to be viewed outside scheduled classes. For the introductory hydrogeology unit, the traditional format with weekly lecture and tutorial classes was maintained, while for the advanced course, weekly 'lectorial' sessions were run for 5 weeks, prior to a 3-day online 'intensive' course, focusing on solving groundwater flow and solute transport modeling problems using MODFLOW-MTD3 (facilitated through online classes). The 'lectorials' involved real-time (online) delivery of lecture content, followed by class discussions and guided, practical problem-solving exercises. Key aspects of the groundwater coursework delivered, and how these were affected by the online 'pivot' are briefly outlined below:

Physical models to demonstrate controls on groundwater flow and solute transport

Prior to the pandemic, benchtop scale models of a heterogeneous layered aquifer system (supplied by Ward's Scientific) were used to demonstrate the behavior of water and solutes, using dyes injected into and extracted from the model at different locations. Such models are effective in demonstrating the effect of changes in hydraulic conductivity and hydraulic gradient on groundwater flow paths, solute movement, and ground–surface water interaction (Rodhe 2012).

During the 2020 mid-year academic break, special permission was given for the instructor to create online content with a professional videographer, using a benchtop hydrogeological model. This resulted in a 60-min video demonstrating concepts normally taught in classes using the model. The video format allowed close-up shots and time-lapse photography (not previously possible), which was considered an additional benefit of the format (Fig. S1). The video was linked to an assessment task which tested how well students understood the key concepts.

Presentations reporting team-based analysis of groundwater data

A key feature of the coursework is an assignment where student teams pick a particular aquifer or region, download data to generate hydrographs showing groundwater level trends, and then conduct their own independent research to identify likely causes of observed changes in groundwater levels through time. Prior to the pandemic, this assignment was assessed at a special student forum held at the end of the semester, with 15-min presentations delivered by each student group and a question-and-answer session. During the pandemic restrictions, the same format was used, but the presentation forum was held online using Microsoft Teams. Facilitated online class sessions using ‘breakout rooms’ allowed group members to interact and discuss their work (as students were not permitted to meet in person) and most groups organized additional out-of-class video calls, document sharing, and messaging group chats to conduct the work.

In both cases (in person and online), students had the chance to critically analyze groundwater data, discuss the possible causes of trends in the data, learn from each other’s analysis and gain confidence speaking about groundwater issues to an audience. The instructor’s reflection was that while the in-person format is generally more engaging and interactive for the students, the online format can still be effective and could be a successful feature of future education in hydrogeology.

Numerical modeling simulations of groundwater flow and solute transport

From 2013 to 2019, students taking the advanced groundwater coursework unit conducted five full-day, in-person groundwater flow and solute transport practical sessions, using MODFLOW-MT3D to develop three-dimensional groundwater flow and solute transport models. The process of building and running these models helped students understand concepts such as steady-state and transient flow, water balances, effects of geological heterogeneity on water flow, inter-aquifer connectivity, and the role of advection, dispersion, and chemical reaction on the development and movement of groundwater contaminant plumes.

During the 2020 pandemic, the five full-day sessions were reduced to three (to avoid ‘digital fatigue’ among students) and supplemented with a series of self-paced video tutorials. The three days of online modeling were conducted with the aid of MS Teams ‘breakout rooms’ to enable small groups to share and discuss their developing models, outputs and answers to questions, and allow the instructor to speak with individuals and small groups to help troubleshoot modeling challenges.

Field and other practical skills: Groundwater level measurement, sampling and hydrogeological observation

Prior to the pandemic, a full-day groundwater field trip was incorporated into the introductory hydrogeology coursework unit each semester. Students traveled to a series of groundwater monitoring sites across a small, coastal, sedimentary aquifer that has historically experienced salt-water intrusion in response to high rates of seasonal groundwater extraction for irrigation (Currell et al. 2013). Students participated in both groundwater level monitoring and groundwater sampling activities.

A key element of the trip is helping students gain observational skills and the ability to develop a working conceptual hydrogeological model of an aquifer, based on observations gained from the field. During 2020, this trip could not run, and students instead conducted desktop-based research into the same groundwater system, reviewing maps, reports, hydrographs and other data made available online. While the students gained valuable experience in conducting desktop hydrogeology assessments, fundamental practical skills (which most students were not otherwise exposed to), could not be taught in 2020. This included groundwater sampling methods, monitoring of physicochemical characteristics of groundwater (during and prior to sample collection) and observing, identifying, and recording important lithological features in soil and aquifer material. In 2021, the field trip was reinstated.

Tests and exams

Prior to the COVID-19 pandemic, the largest component of the assessment was in-person, supervised exams, testing key hydrogeology concepts and quantitative methods. During 2020–2021, these exams were replaced with an online ‘take-home’ examination, which students had a period of 24 h to complete. As found elsewhere in hydrology education (e.g., Fischer and Tatominir 2022) and more broadly in the university sector (Newton and Draper 2025), this online ‘exam’ format, which was common during the pandemic, has been found to have deficiencies in the years since—including an inability to control students’ use of online content that has not been carefully checked or curated, and (more recently) their use of generative artificial intelligence to answer questions without engaging in critical thinking.

Instructor reflections and student feedback

Class attendance and engagement levels

Over the pre-, during-, and post- pandemic years, the instructor found that lecture and class attendance were similar across all years and formats. The presence of online material—often promoted by universities to broaden accessibility to busy students and those studying remotely—had

(in this case) little impact on the overall numbers of students engaging with the material. This was indicated by comparing in-person and online class attendance numbers (as a proportion of total enrolments), and online engagement with digital materials (made possible through embedded analytics tracking unique downloads/views of online material).

In both online and in-person formats, the greatest level of *engagement* (as opposed to just attendance) was attained where students were encouraged to discuss key ideas, case studies and concepts in groups, and contribute these ideas to class discussions in a structured way. The instructor found that this can be achieved either through ‘breakout’ sessions in video call software, or in the traditional classroom; however, in online sessions, engagement among relatively quiet students was far more challenging (e.g., if they chose not to activate their camera or microphone). Social isolation—a major consequence of the pandemic health restrictions—led to more limited student–student interaction during this period, which likely also contributed to poorer overall engagement and learning among certain students (Joshi et al. 2022).

Teaching practical content via an online classroom

Students gave positive overall feedback on their experience of the fully online coursework, giving overall satisfaction ratings of 94.7 and 94.1% for the basic and advanced courses, respectively, and providing comments such as:

“Despite the transition to online learning, I found the course engaging and interactive. I enjoyed the incorporation of case studies in teaching and found that it helped cement how course work is applied in the field and it’s importance.”
“I think the transition to online learning was done incredibly efficiently and the weekly tutes and assignments (provided) the right amount of challeng(e) yet (were) still enjoyable to complete. This course definitely made me feel confident that I’m studying something I want to make into a career!!”

The overall satisfaction rating for the two courses dropped slightly during the 2020 online delivery (from 100% in the two years prior). This may in part relate to ‘online engagement fatigue’ (Maloney et al. 2023); experienced by many students as the pandemic wore on. Qualitative feedback did not indicate any serious concerns about overall quality of the online courses; however, students identified limitations with the fully online modeling tutorial classes:

“I personally found it really hard to do this course online, I think it took me a lot longer to use MODL-FOW and work through issues I was having on my own than it would have if I could physically show someone the issue and have them help me out.”

“This is a course that benefits from in-person teaching, especially with learning the modeling. However, the teacher did a great job with how the breakout rooms were created and the response to burning questions in a considerate and timely matter.”

On reflection, both students and instructor found that the ability to advise students how to build and work with a numerical model in the classroom, and troubleshoot issues directly, was preferable to the online environment, even when screen-sharing, breakout rooms and other advanced features of the video-call software were available.

Field and other practical elements

Multiple students identified the inability to conduct fieldwork as a deficiency of fully online coursework, and some also highlighted the lack of ability to form direct personal connections with their peers as a key challenge:

“It is a shame the tutorials were unable to be completed in person, I know I would have gotten more out of real-life collaboration, especially as the only person I knew pre the move to online learning ended up dropping the course. Likewise, it is a shame we couldn’t complete the fieldwork.”

While some tertiary educators (including the instructor) have experimented with ‘virtual’ field trips for hydrogeology and other earth sciences courses (e.g., Caliskan 2011; Thompson et al. 2022), it is unrealistic to expect virtual experiences to substitute for in-person field training. In the field, direct links are made between observations, processes and theory, and key practical skills are developed ‘by doing’, under the supervision of experienced practitioners.

Analysis of student feedback has identified that most students believe virtual field trips can only complement, not replace, in-person field experiences (Evelpidou et al. 2022). Similar findings have been identified in agricultural and resource management education, among both students and instructors (Joshi et al. 2022). However, as has been noted elsewhere, time, resource and health and safety considerations continue to put pressure on university teaching staff to limit in-field training (Caliskan 2011). Industry forums and professional associations have repeatedly expressed to the instructor that it is vital to maintain solid training in field and practical skills and these views are echoed by many in the earth and environmental sciences community.

Traditional vs. ‘take home’ examinations

While the ‘open book, open Internet’ environment for examination (utilized during 2020 and 2021) has been argued by some educators to be a realistic simulation of the real-world professional environment, the instructor’s experience was that

students typically conducted much less study revision in preparation for these assessments, relying on being able to access information ‘on demand’ during the exam, rather than learning the information and methods in advance, and applying them to problems/scenarios given. This led to relatively poor results in the ‘take home’ exams, and (arguably) a shallower understanding of the material. Some (but not all) tertiary institutions have reinstated timed, in-person, supervised exams as a key element of assessment, particularly in response to concerns over students’ uncontrolled use of generative artificial intelligence (AI), and other forms of cheating. However, Newton and Draper (2025) identified that many universities continue to use online, unsupervised examinations (including over 75% of surveyed universities in the UK during 2023–2024); raising serious questions about quality assurance.

A ‘divide’ in groundwater education during and after COVID-19 restrictions

Successful online education has a number of pre-conditions which must be carefully considered as we look to the future of hydrogeology training in the 21st century. Online learning requires minimum standards of information technology (IT) hardware, software and connectivity. Contrary to common perceptions of university students as well-equipped ‘digital natives’, the instructor’s experience during the pandemic was that many students lacked access either to necessary IT infrastructure and/or stable, disturbance-free learning environments to operate effectively and succeed with online coursework. While university grants were in some cases available to students to assist in meeting basic IT infrastructure requirements, these schemes rarely (if ever) provide support for students to have a stable, disturbance-free working environment.

Based on the instructor’s experience since the pandemic restrictions eased, one of the key impediments to effective education is a divide between ‘haves and have-nots’, the

latter of whom struggle to find adequate space, equipment and study conditions to effectively utilize online materials while completing their coursework. This has been identified as a key aspect of digital inequality (DI), which remains a pressing and un-resolved issue across the tertiary education sector (Czerniewicz 2022; Imran 2023; Hartnett et al. 2023).

Addressing DI and preventing a divide in groundwater education in the years ahead must therefore be a key priority. In the instructor’s experience, traditional classroom-based formats for delivering and assessing content offer a standardized mechanism to ‘level the playing field’ for many aspects of hydrogeology education, and are preferred by most students. Nonetheless, it must be acknowledged that some students seeking to study hydrogeology are unable to attend some or all in-person classes, and mechanisms should be in place to ensure groundwater education does not exclude these people. Initiatives such as The Groundwater Project, which provides freely available online groundwater textbooks in multiple languages, are an important step in this regard. Given that many of the world’s most pressing groundwater problems affect low-income communities (e.g., Dobbin 2020; Balasooriya et al. 2023), resourcing and institutional support must be prioritized to ensure barriers do not prevent members of these communities from training as groundwater professionals.

Conclusions

The COVID-19 pandemic necessitated a rapid and far-reaching transformation of hydrogeology education in 2020, and many changes remain, in the form of greater reliance on online materials and delivery in universities. The experience of designing and delivering groundwater coursework at a major Australian university before, during, and after pandemic lockdowns has important lessons for hydrogeology educators worldwide. Firstly, online material can be effective in helping students grasp key concepts of groundwater and contaminant flow. However, careful planning and curation

Table 1 Summary of the major elements of online hydrogeology education during the COVID-19 pandemic restrictions and their effectiveness/suitability for future education

Elements that worked well	Elements that didn’t work well
‘Lectorial’ sessions where content was delivered by the instructor, and then problems were discussed and solved by students in ‘breakout room’ sessions	‘Virtual’ field trip based on desktop/digital materials only. While desktop studies are effective for learning some key skills and concepts, and are a critical part of the groundwater profession, practical skills are better demonstrated and learned in-person
Oral presentation assessment, where students conduct independent (desktop) research and then present findings of their collaborative work (e.g., analyzing groundwater data available in public portals and digital reports) via video call	Teaching groundwater flow and solute transport modeling through online (video call) software. Students found it challenging to troubleshoot modeling issues online, and suffered ‘digital fatigue’ by the end of the course
Use of high-quality pre-recorded video content to demonstrate concepts of groundwater flow and solute transport (using benchtop model) and then assess understanding of these concepts with questions	Online (unsupervised) examinations to test hydrogeology fundamentals and quantitative skills. The quality of work was generally lower than traditional supervised exams, and assessment integrity was difficult to enforce

of the material is needed, with sufficient time and financial resourcing to ensure quality materials. Secondly, there are aspects of groundwater education, particularly fieldwork and other practical-based skills, that are ill-suited to the online environment. While ‘virtual’ field trips and online classes can complement in-person experiences, the dominant view among the hydrogeology community is that these should not replace hands-on training and in-person interaction. Table 1 summarizes key learnings from the experience, based on student feedback and instructor reflections:

Finally, a key challenge for hydrogeology education in the 21st century is ensuring digital inequality does not hamper opportunities for certain students—particularly those from low-income communities—to participate in and succeed in groundwater courses. Some of the most pressing global groundwater problems occur where students and practitioners have limited access to quality learning infrastructure and resources, and so efforts must continue to ensure opportunities for everyone to contribute to solving the world’s groundwater challenges.

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